

The size of the tongue movement area affects the temporal coordination of consonants and vowels— A proof of concept on investigating speech rhythm

Fabian Tomaschek^{a)}

*Department of General Linguistics, University of Tübingen, Tübingen, Germany
fabian.tomaschek@uni-tuebingen.de*

Adrian Leemann

*Department of Linguistics and English Language, Lancaster University, Lancashire,
England
a.leemann@lancaster.ac.uk*

Abstract: Recent research has revealed substantial between-speaker variation in speech rhythm, which in effect refers to the coordination of consonants and vowels over time. In the current proof-of-concept study, the hypothesis was investigated that these idiosyncrasies arise, in part, from differences in the tongue's movement amplitude. Speech rhythm was parameterized by means of the percentage over which speech is vocalic (%V) in the German pronoun “sie” [zi:]. The findings support the hypothesis: all else being equal, idiosyncratic %V values behaved proportionally to a speaker's tongue movement area. This research underlines the importance of studying language-external factors, such as a speaker's individual tongue movement behavior, to investigate variation in temporal coordination.

© 2018 Acoustical Society of America

[DDO]

Date Received: June 15, 2018 **Date Accepted:** October 16, 2018

1. Introduction

The building blocks of speech rhythm are heavily debated in phonetics—and researchers have provided widely differing definitions of speech rhythm. Ramus *et al.* (1999), for example, have emphasized the importance of variability in consonantal and vocalic intervals for the description of speech rhythm, while O'Dell and Nieminen (2009) model speech rhythm as a coupled oscillator. He (2018), on the other hand, emphasizes the role of syllable intensity in the study of speech rhythm. The temporal coordination of vowels and consonants has been shown to vary as a function of language, dialect, and even sentence as a result of the articulated word material (Arvaniti, 2012; Dellwo, 2010; Dellwo *et al.*, 2012; Ferragne and Pellegrino, 2004; Grabe and Low, 2002; Leemann *et al.*, 2012; Ramus *et al.*, 1999; White and Mattys, 2007a,b; Wiget *et al.*, 2010). Beyond these mainly linguistic domains, there is a large body of research indicating that variability in temporal coordination can also be speaker-specific (He and Dellwo, 2017, 2016; Wiget *et al.*, 2010; Yoon, 2010). These between-speaker differences are robust to changes in language (Dellwo and Fourcin, 2013; Dellwo *et al.*, 2012, 2015; Leemann *et al.*, 2015) and to various sources of within-speaker variability such as speaking styles (spontaneous vs read), speaking rates (*fast* vs *slow*), and changing channels (hifi audio vs telephony) (Dellwo, 2010; Leemann *et al.*, 2015). What remains largely unanswered in these studies is the question of why speakers exhibit such specificity, i.e., idiosyncrasy in temporal coordination.

Dellwo *et al.* (2015) and Leemann *et al.* (2015) have hypothesized that a speaker's anatomical configuration directly results in idiosyncratic movements of the articulators, driving between-speaker variability in temporal coordination. Indeed, a wide body of research shows that a speaker's anatomical makeup affects their articulatory gestures and, as a result, the production of consonants and vowels. For example, speakers with longer vocal tracts exhibit larger displacements during the production of vowels (Fuchs *et al.*, 2008; Winkler *et al.*, 2006). The steepness of the alveolo-palatal region influences articulatory patterns during the production of the [s] and [ʃ] contrast (Weirich and Fuchs, 2006). Speakers with flat palates retract their tongues to produce the contrast, while speakers with steep palates additionally perform an elevation

^{a)} Author to whom correspondence should be addressed.

motion—mirrored by an increase in vertical movement of the tongue. Furthermore, the variability in vowels’ articulatory targets is higher in speakers with a “domed” palate compared to speakers with a flat palate, indicating that a smaller tongue movement area (TMA) results in smaller target locations (Brunner *et al.*, 2009); similar findings have been reported for consonants (Rudy and Yunusova, 2013). Given that anatomical differences are mirrored in changes of articulatory movement amplitudes, which result in changes of the temporal coordination of articulatory gestures (Hertrich and Ackermann, 2000), a link between anatomical differences, movement amplitude, and the temporal coordination of vowels and consonants may be possible. The current paper presents a proof of concept that such a link might exist: we speculate that one particular rhythm metric, the percentage of vocalic intervals in the speech signal (%V), behaves proportionally to the size of the TMA. In other words: we hypothesize that the larger the TMA, the larger the average %V values. To anticipate our results: We indeed find that average %V values are proportional to the size of the TMA for a simple monosyllabic German word. In future studies, we need to show that the current findings can be replicated with more complex syllable structures in larger portions of speech, in spontaneous as well as lab speech.

2. Methods and materials

For the present study, we use articulations recorded in the study of Tomaschek *et al.* (2014, 2018b), and who recorded electromagnetic articulography of 17 native German speakers [nine females, mean age of 25.6, standard deviation (sd)=3] using a Northern Digital Inc. Wave articulograph (sampling frequency of 100 Hz) in a sound-treated booth at the Department of Linguistics of the University of Tübingen. For all experiments, the audio signal was recorded (sampling rate: 22.05 kHz, 16 bit) and synchronized with the articulatory recordings. Tongue movements were captured by three sensors: one slightly behind the tongue tip, one at the tongue middle, and one at the tongue body (distance between each sensor: approximately 1 cm, cf. Fig. 1). A 6D reference sensor, which was used to correct for head movements in an online fashion during the recording of tongue movements, was attached to the speaker’s forehead. In addition, tongue sensor positions were rotated such that their origin was located at the center of a bite plate, to which three sensors were attached in a triangular configuration. The bite plate position was recorded prior to the main experiment. In this way, the back–front direction of the tongue was aligned with the horizontal axis, with more positive horizontal values toward the front of the mouth, and more positive vertical values toward the top of the oral cavity.

We operationalized the tongue body’s vertical and horizontal movement amplitude by TMA, as illustrated in Fig. 1. TMA was operationalized by the area of a rhombus,

$$\text{TMA} = \frac{\Delta_x \times \Delta_y}{2}, \quad (1)$$

where the length of the diagonals, Δ_x and Δ_y , was represented by the differences between the most extreme sensor positions in the vertical and horizontal axes for each subject during the entire recording block, taking into account all articulated material.

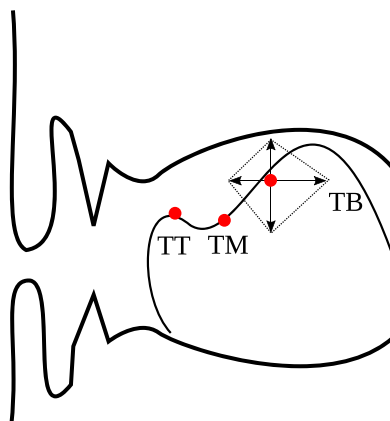


Fig. 1. (Color online) Midsagittal cut through the mouth. The rhombus around the tongue body sensor illustrates the parameterization of TMA.

To exclude an overly strong influence of outliers, we excluded extreme positions larger than 2.5 sds away from the mean.

TMA in the tongue tip and tongue body position had a Spearman-Rank correlation of $\rho = 0.725$, indicating that the tongue body and tongue tip measurements capture the same movement process. We therefore used the TMA of the tongue body sensor. In order to account for changes of TMA across the experiment due to fatigue as well as for sensor location changes of the sensors after reattachment (which happened on average twice per experiment), TMA was assessed in each recording block. To reduce the effect of inconsistencies from collecting data in multiple sessions, the mean TMA across recording blocks was calculated for each participant and used in the analysis.

We expect %V values to be proportional to the individual TMA. On the basis of the work by Dellwo *et al.* (2012) and Leemann *et al.* (2015), who showed speaker-specificity in several rhythm metrics, we investigated how TMA correlates with the variability of the durational proportion of the vowel in the syllable (i.e., %V). As Dellwo *et al.* (2012) and Leemann *et al.* (2015) have shown, this metric reveals most between-speaker differences, more so than rhythm metrics such as VarcoS (syllable-based, rate-normalized standard deviations of vocalic or consonantal interval durations), PVIS (syllable-based, rate-normalized average differences between consecutive vocalic or consonantal interval durations), and intensity peak measures (rate-normalized standard deviations of syllable-peak-to-syllable-peak interval durations).

Although %V is a robust measure, temporal characteristics of words and segments have been repeatedly shown to vary due to intrinsic durations of vowels and consonants (e.g., Antoniadis and Strube, 1984), number of segments and syllables in the word (Altmann, 1980), coarticulation (Magen, 1997; Öhman, 1966), and even lexical measures such as word frequency, which affect articulatory speed and gestural coordination (Gahl, 2008; Tomaschek *et al.*, 2018a, 2018b; Whalen, 1991).

To avoid these sources of variation, we focused on a highly frequent, therefore well-practiced, single monosyllabic word with a CV structure—the German pronoun “sie” [zi:] (*they*), embedded in 378 “sie + verb” phrases. Verbs were canonical bisyllabic German words containing either an [i] or an [a] as stem vowel and a [ə] in the second, unstressed, syllable. Thirty percent of the canonically bisyllabic verbs were articulated with a reduced [ə]. A more detailed description of the material can be found in Tomaschek *et al.* (2018c).

Speakers were instructed to read the “sie+verb” phrases aloud after presentation on a computer screen. The presentation was pseudo-randomized for each participant and divided into three parts. Each part was presented once in a *slow* (odd blocks, inter-stimulus-time: 600 ms; presentation-time: 800 ms) and once in a *fast* speaking rate condition (even blocks, inter-stimulus-time: 300 ms; presentation-time: 450 ms). In total, six blocks of stimuli were presented. Word and phone boundaries were determined by a forced aligner for German (Rapp, 1995) and manually corrected where necessary.

The setup of this study is ideal for investigating correlations between TMA and %V for two reasons: (1) it provides a controlled consonant-vowel sequence, and (2) the lexical and linguistic properties of the following verbs will affect local speaking rate and articulatory amplitude. If idiosyncratic differences in the tongue’s movement amplitude truly correlate with %V, then the correlation should arise in addition to temporal variation resulting from the context of “sie.”

3. Results and discussion

The dependent variable in this analysis was the ratio between the duration of [i:] and the duration of the entire [zi:] word (%V). %V was fit with the following predictors:

- TMA, as explained above;
- word repetition, a control variable to account for the effects on the phonetic characteristics due to repeatedly articulating *sie* during the experiment (Fowler, 1988; Fowler and Housum, 1987);
- syllable rate, gauged by the number of syllables per second on the basis of the entire phrase’s duration, i.e., “sie + verb.” Syllable rate strongly correlated with [zi:] duration (Pearson’s product-moment correlation $\rho = -0.961$), which is why we used only syllable rate during the analysis;
- tangential peak velocity, to control for the potential strategy that speakers with larger TMAs might locally increase their articulatory velocity to counteract their larger movement area, measured in each instance of “sie”;
- speaking rate condition, with the factor levels *fast* or *slow*.

We included speaker, following word, and block number as random intercepts in our analysis. TMA, syllable rate, and peak velocity were log-transformed for the analysis to reduce an overly strong influence of outliers on the regression. The data and analyses can be found in the supplementary material downloadable from <https://osf.io/rck2g/>.

Prior to the regression analysis, we fit %V in a random forest analysis (R package party, number of trees = 500, number of randomly selected variables = 2). The variable importance informed us that TMA is the most important variable for predicting %V, followed by speaking condition, word repetition, and syllable rate. Regressing %V against TMA is therefore well-supported. We used linear mixed-effects regression [Version 1.1–15 (Bates *et al.*, 2014) for R, R Development Core Team (2018), Version 3.3] to fit %V, in which absolute *t*-values larger than 2 are regarded to indicate significant effects. We used exploratory bottom-up and top-down model building strategies and performed model criticism using *F*-tests. We tested interactions between TMA and other predictors but the data did not support any significant interactions between TMA and the other predictors.¹ Once the final model was found, data points with residuals larger than 2.5 were excluded to reduce an overly strong influence from outliers and the model was refit again.

Table 1 presents the results of the final mixed-effects linear regression model. Results are illustrated in Fig. 2. We find that %V is significantly proportional to syllable rate [Fig. 2(a)]. It seems as though a greater speaking tempo is achieved by shortening the consonant rather than the vowel. A Spearman's Rank correlation between syllable rate and consonant and vowel duration indicates this to be true ($\rho_{\text{vowel}} = -0.68$ vs $\rho_{\text{consonant}} = -0.82$). This finding contradicts the results of Gay (1978), who reports a stronger duration reduction for vowels rather than compared to consonants. There are several potential reasons for this difference. Here, a coronal fricative was investigated, whereas Gay (1978) used labial plosives, where there is little room to reduce the duration (see also Kessinger and Blumstein, 1998, who found no difference between vowel and plosive). Also, we focused on a real unstressed word, whereas Gay (1978) focused on a stressed non-word. Word repetition yields a similar effect [Fig. 2(b)], with a stronger effect on the consonant than on the vowel, too ($\rho_{\text{vowel}} = -0.07$ vs $\rho_{\text{consonant}} = -0.1$). This is probably due to fatigue during the experiment. A higher peak velocity is reflected in smaller %V values [Fig. 2(c)], indicating a trade-off between vowel duration and articulatory velocity. %V did not differ systematically between the speaking rate conditions (Table 1, line 4), indicating that %V is robust to changes in global speaking rate. It is, however, not robust to local changes in speaking rate, as indicated above. In line with our hypothesis, %V was proportional to TMA [Fig. 2(d)]: speakers with a larger TMA showed higher values of %V. This is probably due to larger traveling distances in the mouth to reach the vocalic target, rendering the vowel longer on average. Interestingly, variability in TMA between speakers was larger in the *fast* ($\text{sd} = 0.46$) than in the *slow* speaking rate condition ($\text{sd} = 0.31$). This might be due to different strategies to increase speaking rate between speakers and might be linked to differences in the association between TMA and peak velocity in the two speaking rate conditions. Performing separate mixed-effects regression analysis for each of the two conditions, we found that TMA had a significant effect on peak velocity in the *fast* ($\beta = 0.71$, $\text{sd} = 0.21$, $t = 3.33$), but no effect in the *slow* speaking rate condition ($\beta = 0.28$, $\text{sd} = 0.19$, $t = 1.5$). The separation of the two conditions was motivated by a significant interaction between these two predictors in a pilot model.

The question further arises as to whether TMA affects only the duration of segments relative to each other within words—as measured by %V—or whether durations of words themselves are also affected. To explore this question, we regressed word duration against TMA and speaking condition. The random intercept structure

Table 1. Summary table of mixed-effects model fitting %V.

Predictor	Estimate	Std. Error	<i>t</i> value
(Intercept)	−0.10	0.15	−0.65
Syllable Rate	0.08	0.00	34.43
Word Repetition	0.03	0.00	11.05
Speaking Rate Cond— <i>fast</i>	−0.02	0.02	−0.83
Peak Velocity	−0.01	0.00	−4.92
Movement Area	0.08	0.03	2.74

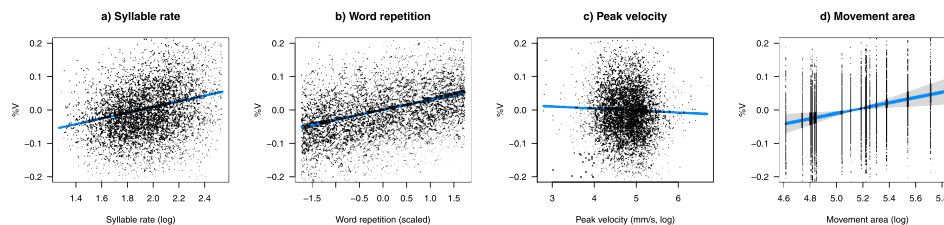


Fig. 2. (Color online) Partial effects in the mixed-effects model. X axis: Predictor. Y -axis: log TMA. Gray dots represent the predicted data.

was the same as in the models reported above. None of the other predictors used in the current study turned out to be significant. TMA failed to be a significant predictor for word duration ($\beta=0.013$, $sd=0.012$, $t=1.019$). %V can be regarded to represent the gestural coordination within words. By contrast, word duration can be interpreted to represent global processes such as gestural orchestration of words or local speaking tempo. The missing effect of TMA on word duration probably indicates that TMA influences the gestural coordination within words, while the overall articulation frame is probably influenced by the external temporal setting of the experiment, as indicated by significantly shorter word duration in the faster speaking condition ($\beta=-0.024$, $sd=0.003$, $t=-7.97$). This effect is probably mediated by the positive correlation between TMA and peak velocity in the *fast* speaking rate condition: A speaker with a large TMA reaches greater articulatory velocities and thereby does not articulate longer words.²

4. General discussion

There is a large body of research showing systematic differences in the temporal variability between speakers regarding the durational intervals of syllables and phones (Dellwo, 2010; Dellwo and Fourcin, 2013; Dellwo *et al.*, 2012, 2015; Leemann *et al.*, 2015; Wiget *et al.*, 2010; Yoon, 2010). In the present paper, we link this temporal variability at the phone level to systematic differences in the TMA of a speaker, i.e., the area in the mid-sagittal plane a tongue covers during articulation. The larger the TMA, the longer the vowel in a monosyllabic isolated word relative to the duration of the word (%V). The data in the current study stems from a temporally well-controlled experiment with fixed articulatory windows. One might hypothesize that speakers with larger TMAs would increase their articulatory speeds—as gauged by peak velocities—to compensate for the additional time it takes for their tongues to travel the longer distances. However, such an effect failed to be recognized. Rather, speakers with larger movement areas did not excel at articulatory speed, indicating that they maintained the same level of articulatory effort as that of speakers with smaller movement areas (cf. Zipf, 1935). As a result, speakers with larger movement areas took longer to articulate vowels, producing acoustically longer vowels.

Given the proof-of-concept nature of the current study, two issues arise. The first issue is centered on the fact that %V is one of many metrics typically used to gauge speech rhythm across large portions of speech containing different syllable structures (Dellwo *et al.*, 2012; Leemann *et al.*, 2015). In order to keep the variance low, we focused on an isolated word with a simple CV structure in the present study. In future studies, we need to show that the current findings can be replicated with more complex syllable structures in larger portions of speech, and in spontaneous as well as lab speech. We furthermore need to investigate other metrics of rhythm (Dellwo *et al.*, 2015; Leemann *et al.*, 2015). Importantly, we regard our findings to indicate idiosyncratic variability above and beyond sources of rhythmic variability such as language and dialect (Grabe and Low, 2002; Ramus *et al.*, 1999; White and Mattys, 2007b).

A second issue that arose, and which will need to be attended to in future studies, is the need to address the nature of the TMA. On the basis of studies linking articulatory variability to differences in oral physiology (Brunner *et al.*, 2009; Fuchs *et al.*, 2008; Rudy and Yunusova, 2013; Weirich and Fuchs, 2006; Winkler *et al.*, 2006), we hypothesize that TMA is proportional to the size of the speaker's oral cavity, which is the true source of idiosyncratic differences in temporal variability. Our findings underline the sensitivity of temporal variability to language-external factors, such as the anatomical make-up of speakers – not only the size of their oral cavity but also individual differences in palate shape. The question as to which factors, language-internal (e.g., language or dialect-related) or language-external, play the most significant role in explaining variability in speech rhythm remains to be determined by future work.

Acknowledgments

This research was funded in part by the Alexander von Humboldt professorship awarded to R. H. Baayen (Grant No. 1141527), and in part by a collaborative grant from the Deutsche Forschungsgemeinschaft (Grant No. BA 3080/3-1).

References and links

¹In pilot analyses, we saw a significant TMA $\times$ speaking rate condition interaction. The significance of this interaction turned out to be the result of enhancement due to collinearity with syllable rate, as correctly observed by an anonymous reviewer (see also Tomaschek *et al.*, 2018c).

²We are thankful to an anonymous reviewer for this suggestion.

- Altmann, G. (1980). "Prolegomena to menzerath's law," *Glottometrika* **2**, 1–10.
- Antoniadis, Z., and Strube, H. (1984). "Untersuchungen zur spezifischen dauer deutscher vokale" ("Investigation of the specific duration of German vowels"), *Phonetica* **41**, 72–87.
- Arvaniti, A. (2012). "The usefulness of metrics in the quantification of speech rhythm," *J. Phonetics* **40**(3), 351–373.
- Bates, D., Maechler, M., Bolker, B., and Walker, S. (2014). lme4: Linear mixed-effects models using Eigen and S4, <http://CRAN.R-project.org/package=lme4> (Last viewed October 30, 2018).
- Brunner, J., Fuchs, S., and Perrier, P. (2009). "On the relationship between palate shape and articulatory behavior," *J. Acoust. Soc. Am.* **125**(6), 3936–3949.
- Dellwo, V. (2010). *Influences of Speech Rate on the Acoustic Correlates of Speech Rhythm: An Experimental Phonetic Study Based on Acoustic and Perceptual Evidence* (Rheinischen Friedrich-Wilhelms-Universität, Bonn, Germany).
- Dellwo, V., and Fourcin, A. (2013). "Rhythmic characteristics of voice between and within languages," *Travaux neuchâtelois de linguistique* **59**, 87–107.
- Dellwo, V., Leemann, A., and Kolly, M.-J. (2012). "Speaker idiosyncratic rhythmic features in the speech signal," in *13th Annual Conference of the International Speech Communication Association 2012*, Vol. 2.
- Dellwo, V., Leemann, A., and Kolly, M.-J. (2015). "Rhythmic variability between speakers: Articulatory, prosodic and lexical factors," *J. Acoust. Soc. Am.* **137**(3), 1513–1528.
- Ferragne, E., and Pellegrino, F. (2004). "Rhythm in read British English: Interdialect variability," in *Proceedings of Interspeech 2004*, Jeju Island, Korea (October 4–8).
- Fowler, C. A. (1988). "Differential shortening of repeated content words produced in various communicative contexts," *Lang. Speech* **31**(4), 307–319.
- Fowler, C. A., and Housum, J. (1987). "Talkers' signaling of 'new' and 'old' words in speech and listeners' perception and use of the distinction," *J. Memory Lang.* **26**(5), 489–504.
- Fuchs, S., Winkler, R., and Perrier, P. (2008). "Do speakers' vocal tract geometries shape their articulatory vowel space?," in *8th International Seminar on Speech Production*, pp. 333–336.
- Gahl, S. (2008). "Thyme and time are not homophones: The effect of lemma frequency on word. Word durations in spontaneous speech," *Language* **84**(3), 474–496.
- Gay, T. (1978). "Effect of speaking rate on vowel formant movements," *J. Acoust. Soc. Am.* **63**(1), 223–230.
- Grabe, E., and Low, E. L. (2002). "Duration variability in speech and the rhythm class hypothesis," *Lab. Phonology* **7**, 515–546.
- He, L. (2018). "Development of speech rhythm in first language: The role of syllable intensity variability," *J. Acoust. Soc. Am.* **143**(6), EL463–EL467.
- He, L., and Dellwo, V. (2016). "The role of syllable intensity in between-speaker rhythmic variability," *Int. J. Speech, Lang. Law* **23**(2), 243–273.
- He, L., and Dellwo, V. (2017). "Between-speaker variability in temporal organizations of intensity contours," *J. Acoust. Soc. Am.* **141**(5), EL488–EL949.
- Hertrich, I., and Ackermann, H. (2000). "Lip-jaw and tongue-jaw coordination during rate-controlled syllable repetitions," *J. Acoust. Soc. Am.* **107**(4), 2236–2247.
- Kessinger, R. H., and Blumstein, S. E. (1998). "Effects of speaking rate on voice-onset time and vowel production: Some implications for perception studies," *J. Phonetics* **26**(2), 117–128.
- Leemann, A., Dellwo, V., Kolly, M.-J., and Schmid, S. (2012). "Rhythmic variability in Swiss German dialects," in *6th International Conference on Speech Prosody*, Shanghai (May 22–25, 2012), pp. 607–610.
- Leemann, A., Kolly, M.-J., and Dellwo, V. (2015). "Speaker-individuality in suprasegmental temporal features: Implications for forensic voice comparison," *Forensic Sci. Int.* **238C**, 59–67.
- Magen, H. S. (1997). "The extent of vowel-to-vowel coarticulation in English," *J. Phonetics* **25**, 187–205.
- O'Dell, M. L., and Nieminen, T. (2009). "Coupled oscillator model for speech timing: Overview and examples," in *Proceedings of the Xth Conference*, Helsinki, pp. 179–190.
- Öhman, S. (1966). "Coarticulation in VCV utterances: Spectrographic measurements," *J. Acoust. Soc. Am.* **39**(151), 151–168.
- Ramus, F., Nespore, M., and Mehler, J. (1999). "Correlates of linguistic rhythm in the speech signal," *Cognition* **73**(3), 265–292.
- Rapp, S. (1995). "Automatic phonemic transcription and linguistic annotation from known text with hidden Markov models/an aligner for German," in *Proceedings of ELSNET Goes East and IMACS Workshop*, Moscow, Russia.
- R Development Core Team (2018). *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria, <http://www.R-project.org>.
- Rudy, K., and Yunusova, Y. (2013). "The effect of anatomic factors on tongue position variability during consonants," *J. Speech Lang. Hear. Res.* **56**, 137–149.

- Tomaschek, F., Arnold, D., Bröker, F., and Baayen, R. H. (2018a). "Lexical frequency co-determines the speed-curvature relation in articulation," *J. Phonetics* **68**, 103–116.
- Tomaschek, F., Hendrix, P., and Baayen, R. H. (2018c). "Strategies for managing collinearity in multivariate linguistic data," *J. Phonetics* **71**, 249–267.
- Tomaschek, F., Tucker, B. V., Baayen, R. H., and Fasiolo, M. (2018b). "Practice makes perfect: The consequences of lexical proficiency for articulation," *Linguistic Vanguard* **4**(S2), 1–13.
- Tomaschek, F., Tucker, B. V., Wieling, M., and Baayen, R. H. (2014). "Vowel articulation affected by word frequency," in *Proceedings of the 10th ISSP*, Cologne, pp. 425–428.
- Weirich, M., and Fuchs, S. (2006). "Palatal morphology can influence speaker-specific realizations of phonemic contrasts," *J. Speech, Lang., Hear. Res.* **56**, 1894–1908.
- Whalen, D. H. (1991). "Infrequent words are longer in duration than frequent words," *J. Acoust. Soc. Am.* **90**(4), 2311–2311.
- White, L., and Mattys, L. S. (2007a). "Calibrating rhythm: First language and second language studies," *J. Phonetics* **35**(4), 501–522.
- White, L., and Mattys, L. S. (2007b). "Rhythmic typology and variation in first and second languages," *Linguistic Theory* **282**, 237–257.
- Wiget, L., White, L., Schuppler, B., Grenon, I., Rauch, O., and Mattys, S. L. (2010). "How stable are acoustic metrics of contrastive speech rhythm?," *J. Acoust. Soc. Am.* **127**(3), 1559–1569.
- Winkler, R., Fuchs, S., and Perrier, P. (2006). "The relation between differences in vocal tract geometry and articulatory control strategies in the production of French vowels: Evidence from MRI and modeling," in *Proceedings of the 7th International Seminar on Speech Production (ISSP)*, Ubatuba-SP, Brazil, pp. 509–516.
- Yoon, T.-J. (2010). "Capturing inter-speaker invariance using statistical measures of rhythm," in *Speech Prosody 2010*, Illinois.
- Zipf, G. (1935). *The Psycho-Biology of Language. An Introduction to Dynamic Philology* (MIT Press, Cambridge), pp. xxii, 625 pp.